

Substitution method



1 Solve the following simultaneous equations:

a $y = 6x + 56$
 $y = 2x + 24$

c $y = 5x + 37$
 $5y = 35x + 265$

e $y = 3x + 14$
 $x + y = 2$

g $y = -6x + 59$
 $-2x + y = -13$

i $y = -3x + 9$
 $6x - y = 0$

k $y = -3x - 23$
 $x + 4y = -37$

m $5x - 2y = -10$
 $y = -5x + 3$

b $y = -5x - 22$
 $y = 7x + 38$

d $y = -4x - 27$
 $2y = 10x + 54$

f $y = -3x + 22$
 $x + y = 10$

h $y = 2x + 13$
 $x - y = -11$

j $y = 4x - 28$
 $3x + 5y = 44$

l $y = 2x - 15$
 $3x - 4y = 20$

n $2x + 3y = 26$
 $8x - 5y = 36$

2 Consider the given system of equations:

$$\begin{aligned} 3x - 7y &= 4 \\ -12x + 28y &= -16 \end{aligned}$$

- a** Rearrange $3x - 7y = 4$ to find x in terms of y .
- b** Substitute your expression for x into $-12x + 28y = -16$ and solve for y .
- c** State whether the system of equations is inconsistent, dependent or independent.

3 Consider the given system of equations:

$$\begin{aligned} 5x + 3y &= -64 \\ 7x + 4y &= -88 \end{aligned}$$

- a** Rearrange $5x + 3y = -64$ to make y the subject in terms of x .
- b** Solve for x .
- c** Hence, solve for y .

4 Solve the following systems of equations:



a $2x + 3y = 26$
 $8x - 5y = 36$

b $2x + 7y = 23$
 $5x + 6y = 0$

c $5x + 4y = 10$
 $3x + 7y = -17$

d $4x + 3y = 17$
 $9x - 4y = 6$

e $-7f + 2g = -\frac{2}{5}$
 $-21f + 10g = \frac{18}{5}$

f $-7p - 2q = -\frac{99}{40}$
 $-21p - 10q = -\frac{85}{8}$

g $-4a + 7b = \frac{23}{2}$
 $-20a + 21b = \frac{45}{2}$

h $4v - 7t = -\frac{143}{20}$
 $12v - 35t = -\frac{779}{20}$

5 Consider the following system of linear equations:

$$-6x - 2y = -28$$

$$2x + 16y = 40$$

$$4x - 2y = 12$$

- a Find the values of x and y that satisfy the first two equations.
- b Determine if this solution satisfies the third equation.
- c Hence state whether the lines are concurrent.

Applications

6 The length of a rectangle measures 10 units more than the width, and the perimeter of the rectangle is 56 units. Let y be the width and x be the length of the rectangle.

- a Consider that the length of the rectangle is 10 units more than the width to set up Equation 1.
- b Consider that the perimeter of the rectangle is equal to 56 to set up Equation 2.
- c Solve for y to find the width.
- d Hence, solve for x to find the length.

7 There are 22 members in a group and the men outnumber the women by 6. Let x and y be the number of women and men in the group respectively.

- a Consider that the men outnumber the women by 6 to set up Equation 1.
- b Consider that there are a total of 22 members in the group to set up Equation 2.
- c Solve for x to find the number of women in the group.
- d Hence, solve for y to find the number of men in the group.

- 8 A mother is currently 10 times older than her son. In two years time, she will be 7 times older than her son. Let x and y be the present ages of the son and mother respectively.
- Set up two equations that describe the statements above.
 - Solve for x to find the son's current age.
 - Hence, solve for y to find the mother's current age.
- 9 A man is 5 times as old as his son. Four years ago the man was 9 times as old as his son. Let x and y be the ages of the man and his son respectively.
- Set up two equations that describe the statements above.
 - Solve for y to find the age of the son.
 - Hence, solve for x to find the age of the man.
- 10 Valerie has \$3000 to invest, and wants to split it up between two accounts. Account A earns 6% annual interest and Account B earns 8% annual interest. Her target is to earn \$204 total interest from the two accounts in one year. Let x and y be the amounts, in dollars, that she invests in accounts A and B respectively.
- Set up two equations that describe the statements above.
 - Solve for x and y .
- 11 Neil bought some fresh produce. He picked up 2 apples, and 3 kiwi fruit. The cost of Neil's shopping was \$10.08. Eileen went to the same shop and bought 5 apples and 7 kiwi fruit. The cost of Eileen's shopping was \$24.08. Let f and g be the price of apples and kiwi fruit respectively.
- Set up two equations that describe the statements above.
 - Find the price of apples and kiwi fruit respectively.
- 12 The number of new jobs created in Venhurst varies greatly each year. The number of jobs created in 2013 was 190 000 less than quadruple the number of jobs created in 2008. This is equivalent to an increase of 560 000 jobs created from 2008 to 2013. Let x be the number of jobs created in 2008 and let y be the number of jobs created in 2013.
- Set up two equations that describe the statements above.
 - Find the amount of jobs created in 2008 and 2013.
- 13 A taxi driver charges \$ x for the initial pick up, and then \$ y per km travelled. A journey of 10km costs \$105, and a journey of 15km costs \$155.
- Set up two equations that describe the statements above.
 - Find the amount of the fixed charge and the charge per km.



- 14 The coach of a tennis team buys 7 rackets and 6 balls for \$3800. Later, he buys 3 rackets and 5 balls for \$1750. Find the price of rackets and balls respectively.
- 15 Toby's piggy bank contains only 5¢ and 10¢ coins. If it contains 60 coins with a total value of \$3.75, find the number of each type of coin.
- 16 Michael sells hot dogs and soda. Each hotdog costs \$1.50 and each soda costs \$0.50. At the end of the day he made a total of \$78.50 for a total of 87 hot dogs and sodas combined. How many hotdogs and sodas were sold?